**VIT-AP UNIVERSITY, ANDHRA PRADESH**

**Lab Sheet 6: MongoDB Basic commands**

**Branch/ Class:** B.Tech/M.Tech **Date: 26/2/26**
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1. Use MongoDB to implement the following DB operations

   1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".
   2. Write a MongoDB query to display all the databases.
   3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.
   4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.
   5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.
   6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.
   7. Write a MongoDB query to display only vehicle name and price in all the collection of the database
   8. Write a MongoDB query to display two_wheelers from a particular company
   9. Write a MongoDB query to display four_wheelers available in diesel variants
   10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

# Output:

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
admin> use vehicles
switched to db vehicles
vehicles> show dbs
admin   40.00 KiB
config  60.00 KiB
local   40.00 KiB
vehicles> db.createCollection("two_wheelers", {
|   capped: true,
|   size: 1048576,
|   max: 100
| })
{ ok: 1 }
vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
vehicles> db.two_wheelers.insertMany([
|   {
|     bike_name: "Honda Activa 6G",
|     model: "gearless",
|     category: "125cc",
|     colors_available: ["Pearl Precious White", "Rebel Red Metallic", "Mat Axis Grey Metallic", "Black"],
|     manufacturer: "Honda",
|     performance: 8,
|     timestamp: ISODate("2020-06-01"),
|     price: 78000
|   },
|   {
|     bike_name: "Yamaha MT 15 V2",
|     model: "gear",
|     category: "150cc",
|     colors_available: ["Dark Matte Blue", "Metallic Black", "Racing Blue", "Cyan Storm"],
|     manufacturer: "Yamaha",
|     performance: 9,
|     timestamp: ISODate("2021-09-15"),
|     price: 169000
|   },
|   {
|     bike_name: "Bajaj Pulsar NS200",
|     model: "gear",
|     category: "200cc",
|     colors_available: ["Pearl Metallic White", "Burnt Red", "Graphite Black"],
|     manufacturer: "Bajaj",
|     performance: 8.5,
|     timestamp: ISODate("2012-07-01"),
|     price: 158000
|   },
|   {
|     bike_name: "TVS Jupiter 125",
|     model: "gearless",
|     category: "125cc",
|     colors_available: ["Prismatic Red", "Metallic Blue", "Ivory Brown"],
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
|     bike_name: "TVS Jupiter 125",
|     model: "gearless",
|     category: "125cc",
|     colors_available: ["Prismatic Red", "Metallic Blue", "Ivory Brown"],
|     manufacturer: "TVS",
|     performance: 7.5,
|     timestamp: ISODate("2021-11-01"),
|     price: 85000
|   },
|   {
|     bike_name: "Royal Enfield Classic 350",
|     model: "gear",
|     category: "350cc",
|     colors_available: ["Halcyon Green", "Gunmetal Grey", "Dark Stealth Black"],
|     manufacturer: "Royal Enfield",
|     performance: 7,
|     timestamp: ISODate("2021-09-01"),
|     price: 195000
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fcf018166c492788563b1'),
    '1': ObjectId('699fcf018166c492788563b2'),
    '2': ObjectId('699fcf018166c492788563b3'),
    '3': ObjectId('699fcf018166c492788563b4'),
    '4': ObjectId('699fcf018166c492788563b5')
  }
}
vehicles> db.four_wheelers.insertMany([
|   {
|     vehicle_name: "Maruti Suzuki Swift",
|     model: "own",
|     category: "car",
|     variants: ["LXi", "VXi", "ZXi", "ZXi+", "AMT"],
|     manufacturer: "Maruti Suzuki",
|     performance: 8,
|     timestamp: ISODate("2024-05-01"),
|     price: 675000
|   },
|   {
|     vehicle_name: "Tata Nexon",
|     model: "own",
|     category: "car",
|     variants: ["Smart", "Pure", "Creative", "Fearless", "Diesel", "AMT"],
|     manufacturer: "Tata Motors",
|     performance: 8.5,
|     timestamp: ISODate("2023-09-14"),
|     price: 850000
|   },
```

mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
|   },
|   {
|     vehicle_name: "Mahindra Bolero",
|     model: "own",
|     category: "car",
|     variants: ["B4", "B6", "B6 Opt", "Diesel"],
|     manufacturer: "Mahindra",
|     performance: 7,
|     timestamp: ISODate("2000-01-01"),
|     price: 950000
|   },
|   {
|     vehicle_name: "Eicher Pro 2049",
|     model: "commercial",
|     category: "mini truck",
|     variants: ["17 Feet", "19 Feet", "Diesel"],
|     manufacturer: "Eicher",
|     performance: 6.5,
|     timestamp: ISODate("2018-04-01"),
|     price: 1650000
|   },
|   {
|     vehicle_name: "Tata Ace Gold",
|     model: "commercial",
|     category: "mini truck",
|     variants: ["Petrol", "CNG", "Diesel", "GVW 1510 kg"],
|     manufacturer: "Tata Motors",
|     performance: 7,
|     timestamp: ISODate("2019-02-01"),
|     price: 595000
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fcf338166c492788563b6'),
    '1': ObjectId('699fcf338166c492788563b7'),
    '2': ObjectId('699fcf338166c492788563b8'),
    '3': ObjectId('699fcf338166c492788563b9'),
    '4': ObjectId('699fcf338166c492788563ba')
  }
}
vehicles> db.two_wheelers.find().pretty()
| db.four_wheelers.find().pretty()
[
  {
    _id: ObjectId('699fcf338166c492788563b6'),
    vehicle_name: 'Maruti Suzuki Swift',
    model: 'own',
    category: 'car',
    variants: [ 'LXi', 'VXi', 'ZXi', 'ZXi+', 'AMT' ],
    manufacturer: 'Maruti Suzuki',
```

mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
    manufacturer: 'Maruti Suzuki',
    performance: 8,
    timestamp: ISODate('2024-05-01T00:00:00.000Z'),
    price: 675000
  },
  {
    _id: ObjectId('699fcf338166c492788563b7'),
    vehicle_name: 'Tata Nexon',
    model: 'own',
    category: 'car',
    variants: [ 'Smart', 'Pure', 'Creative', 'Fearless', 'Diesel', 'AMT' ],
    manufacturer: 'Tata Motors',
    performance: 8.5,
    timestamp: ISODate('2023-09-14T00:00:00.000Z'),
    price: 850000
  },
  {
    _id: ObjectId('699fcf338166c492788563b8'),
    vehicle_name: 'Mahindra Bolero',
    model: 'own',
    category: 'car',
    variants: [ 'B4', 'B6', 'B6 Opt', 'Diesel' ],
    manufacturer: 'Mahindra',
    performance: 7,
    timestamp: ISODate('2000-01-01T00:00:00.000Z'),
    price: 950000
  },
  {
    _id: ObjectId('699fcf338166c492788563b9'),
    vehicle_name: 'Eicher Pro 2049',
    model: 'commercial',
    category: 'mini truck',
    variants: [ '17 Feet', '19 Feet', 'Diesel' ],
    manufacturer: 'Eicher',
    performance: 6.5,
    timestamp: ISODate('2018-04-01T00:00:00.000Z'),
    price: 1650000
  },
  {
    _id: ObjectId('699fcf338166c492788563ba'),
    vehicle_name: 'Tata Ace Gold',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'Petrol', 'CNG', 'Diesel', 'GVW 1510 kg' ],
    manufacturer: 'Tata Motors',
    performance: 7,
    timestamp: ISODate('2019-02-01T00:00:00.000Z'),
    price: 595000
  }
]
vehicles> db.getCollectionNames().forEach(function(coll) {
|   print("\nCollection: " + coll);
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

vehicles> db.getCollectionNames().forEach(function(coll) {
|   print("\nCollection: " + coll);
|   db[coll].find({}, {
|     bike_name: 1,
|     vehicle_name: 1,
|     price: 1,
|     _id: 0
|   }).forEach(printjson);
| })

Collection: two_wheelers
{
  bike_name: 'Honda Activa 6G',
  price: 78000
}
{
  bike_name: 'Yamaha MT 15 V2',
  price: 169000
}
{
  bike_name: 'Bajaj Pulsar NS200',
  price: 158000
}
{
  bike_name: 'TVS Jupiter 125',
  price: 85000
}
{
  bike_name: 'Royal Enfield Classic 350',
  price: 195000
}

Collection: four_wheelers
{
  vehicle_name: 'Maruti Suzuki Swift',
  price: 675000
}
{
  vehicle_name: 'Tata Nexon',
  price: 850000
}
{
  vehicle_name: 'Mahindra Bolero',
  price: 950000
}
{
  vehicle_name: 'Eicher Pro 2049',
  price: 1650000
}
{
  vehicle_name: 'Tata Ace Gold',
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

  vehicle_name: 'Tata Ace Gold',
  price: 595000
}

vehicles> db.two_wheelers.find({ manufacturer: "Honda" }).pretty()
[
  {
    _id: ObjectId('699fcf018166c492788563b1'),
    bike_name: 'Honda Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [
      'Pearl Precious White',
      'Rebel Red Metallic',
      'Mat Axis Grey Metallic',
      'Black'
    ],
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

    _id: ObjectId('699fcf338166c492788563b9'),
    vehicle_name: 'Eicher Pro 2049',
    model: 'commercial',
    category: 'mini truck',
    variants: [ '17 Feet', '19 Feet', 'Diesel' ],
    manufacturer: 'Eicher',
    performance: 6.5,
    timestamp: ISODate('2018-04-01T00:00:00.000Z'),
    price: 1650000
  },
  {
    _id: ObjectId('699fcf338166c492788563ba'),
    vehicle_name: 'Tata Ace Gold',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'Petrol', 'CNG', 'Diesel', 'GVW 1510 kg' ],
    manufacturer: 'Tata Motors',
    performance: 7,
    timestamp: ISODate('2019-02-01T00:00:00.000Z'),
    price: 595000
  }
]
vehicles> db.two_wheelers.find(
|   { performance: { $gt: 5 } },
|   {
|     bike_name: 1,
|     category: 1,
|     manufacturer: 1,
|     performance: 1,
|     _id: 0
|   }
| ).pretty()
[
  {
    bike_name: 'Honda Activa 6G',
    category: '125cc',
    manufacturer: 'Honda',
    performance: 8
  },
  {
    bike_name: 'Yamaha MT 15 V2',
    category: '150cc',
    manufacturer: 'Yamaha',
    performance: 9
  },
  {
    bike_name: 'Bajaj Pulsar NS200',
    category: '200cc',
    manufacturer: 'Bajaj',
    performance: 8.5
  },
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
  },
  {
    bike_name: 'TVS Jupiter 125',
    category: '125cc',
    manufacturer: 'TVS',
    performance: 7.5
  },
  {
    bike_name: 'Royal Enfield Classic 350',
    category: '350cc',
    manufacturer: 'Royal Enfield',
    performance: 7
  }
]
vehicles> db.four_wheelers.find(
|   { performance: { $gt: 5 } },
|   {
|     vehicle_name: 1,
|     category: 1,
|     manufacturer: 1,
|     performance: 1,
|     _id: 0
|   }
| ).pretty()
[
  {
    vehicle_name: 'Maruti Suzuki Swift',
    category: 'car',
    manufacturer: 'Maruti Suzuki',
    performance: 8
  },
  {
    vehicle_name: 'Tata Nexon',
    category: 'car',
    manufacturer: 'Tata Motors',
    performance: 8.5
  },
  {
    vehicle_name: 'Mahindra Bolero',
    category: 'car',
    manufacturer: 'Mahindra',
    performance: 7
  },
  {
    vehicle_name: 'Eicher Pro 2049',
    category: 'mini truck',
    manufacturer: 'Eicher',
    performance: 6.5
  },
  {
    vehicle_name: 'Tata Ace Gold',
    category: 'mini truck',
```

```
  },
  {
    vehicle_name: 'Tata Ace Gold',
    category: 'mini truck',
    manufacturer: 'Tata Motors',
    performance: 7
  }
]
vehicles>
```

2. Use MongoDB to implement the following DB operations for a Zoo

    1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.
    2. Write a MongoDB query to display all the databases.
    3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.
    4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.
    5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female),

favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.
7. Write a MongoDB query to display only animal name and expenses in all the collection of the database
8. Write a MongoDB query to display domestic_animals whose life is a particular year
9. Write a MongoDB query to display wild_animals available under a particular care_taker
10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
vehicles> use animal
switched to db animal
animal> show dbs
admin     40.00 KiB
config    84.00 KiB
local     40.00 KiB
vehicles  80.00 KiB
animal> db.createCollection("wild_animals", {
|   capped: true,
|   size: 1048576,
|   max: 100
| })
{ ok: 1 }
animal> db.createCollection("domestic_animals")
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
|     life_span: 25,
|     timestamp: new Date(),
|     expenses: 65000
|   },
|   {
|     animal_name: "Bear",
|     nature: "harm",
|     favorite_foods: ["fish", "honey", "berries"],
|     care_taker_name: "Priya",
|     life_span: 20,
|     timestamp: new Date(),
|     expenses: 78000
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd4eb8166c492788563bb'),
    '1': ObjectId('699fd4eb8166c492788563bc'),
    '2': ObjectId('699fd4eb8166c492788563bd'),
    '3': ObjectId('699fd4eb8166c492788563be'),
    '4': ObjectId('699fd4eb8166c492788563bf')
  }
}
animal> db.domestic_animals.insertMany([
|   {
|     animal_name: "Dog",
|     gender: "male",
|     favorite_foods: ["meat", "chicken", "biscuits"],
|     animal_petname: "Bruno",
|     life_span: 12,
|     timestamp: new Date(),
|     expenses: 25000
|   },
|   {
|     animal_name: "Cat",
|     gender: "female",
|     favorite_foods: ["fish", "milk", "chicken"],
|     animal_petname: "Whiskers",
|     life_span: 15,
|     timestamp: new Date(),
|     expenses: 18000
|   },
|   {
|     animal_name: "Cow",
|     gender: "female",
|     favorite_foods: ["grass", "hay", "grains"],
|     animal_petname: "Lakshmi",
|     life_span: 18,
|     timestamp: new Date(),
|     expenses: 40000
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
|   },
|   {
|     animal_name: "Parrot",
|     gender: "male",
|     favorite_foods: ["seeds", "fruits", "nuts"],
|     animal_petname: "Mithu",
|     life_span: 30,
|     timestamp: new Date(),
|     expenses: 12000
|   },
|   {
|     animal_name: "Rabbit",
|     gender: "female",
|     favorite_foods: ["carrots", "greens", "hay"],
|     animal_petname: "Snowy",
|     life_span: 8,
|     timestamp: new Date(),
|     expenses: 8000
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fd4f18166c492788563c0'),
    '1': ObjectId('699fd4f18166c492788563c1'),
    '2': ObjectId('699fd4f18166c492788563c2'),
    '3': ObjectId('699fd4f18166c492788563c3'),
    '4': ObjectId('699fd4f18166c492788563c4')
  }
}
animal> db.wild_animals.find().pretty()
| db.domestic_animals.find().pretty()
[
  {
    _id: ObjectId('699fd4f18166c492788563c0'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'chicken', 'biscuits' ],
    animal_petname: 'Bruno',
    life_span: 12,
    timestamp: ISODate('2026-02-26T05:06:57.053Z'),
    expenses: 25000
  },
  {
    _id: ObjectId('699fd4f18166c492788563c1'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk', 'chicken' ],
    animal_petname: 'Whiskers',
    life_span: 15,
    timestamp: ISODate('2026-02-26T05:06:57.053Z'),
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
    expenses: 18000
  },
  {
    _id: ObjectId('699fd4f18166c492788563c2'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'hay', 'grains' ],
    animal_petname: 'Lakshmi',
    life_span: 18,
    timestamp: ISODate('2026-02-26T05:06:57.053Z'),
    expenses: 40000
  },
  {
    _id: ObjectId('699fd4f18166c492788563c3'),
    animal_name: 'Parrot',
    gender: 'male',
    favorite_foods: [ 'seeds', 'fruits', 'nuts' ],
    animal_petname: 'Mithu',
    life_span: 30,
    timestamp: ISODate('2026-02-26T05:06:57.053Z'),
    expenses: 12000
  },
  {
    _id: ObjectId('699fd4f18166c492788563c4'),
    animal_name: 'Rabbit',
    gender: 'female',
    favorite_foods: [ 'carrots', 'greens', 'hay' ],
    animal_petname: 'Snowy',
    life_span: 8,
    timestamp: ISODate('2026-02-26T05:06:57.053Z'),
    expenses: 8000
  }
]
animal> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
| db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
[
  { animal_name: 'Dog', expenses: 25000 },
  { animal_name: 'Cat', expenses: 18000 },
  { animal_name: 'Cow', expenses: 40000 },
  { animal_name: 'Parrot', expenses: 12000 },
  { animal_name: 'Rabbit', expenses: 8000 }
]
animal> db.domestic_animals.find({ life_span: 15 }).pretty()
[
  {
    _id: ObjectId('699fd4f18166c492788563c1'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk', 'chicken' ],
    animal_petname: 'Whiskers',
    life_span: 15,
```

mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
    timestamp: ISODate('2026-02-26T05:06:57.053Z'),
    expenses: 18000
  }
]
animal> db.wild_animals.find({ care_taker_name: "Rajesh" }).pretty()
[
  {
    _id: ObjectId('699fd4eb8166c492788563bb'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'antelope', 'zebra' ],
    care_taker_name: 'Rajesh',
    life_span: 12,
    timestamp: ISODate('2026-02-26T05:06:51.864Z'),
    expenses: 85000
  },
  {
    _id: ObjectId('699fd4eb8166c492788563bd'),
    animal_name: 'Elephant',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'fruits', 'leaves' ],
    care_taker_name: 'Rajesh',
    life_span: 60,
    timestamp: ISODate('2026-02-26T05:06:51.864Z'),
    expenses: 150000
  }
]
animal> db.wild_animals.find(
|   { life_span: { $gt: 5 } },
|   { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
| ).pretty()
[
  {
    animal_name: 'Lion',
    favorite_foods: [ 'meat', 'antelope', 'zebra' ],
    expenses: 85000
  },
  {
    animal_name: 'Tiger',
    favorite_foods: [ 'meat', 'deer', 'boar' ],
    expenses: 92000
  },
  {
    animal_name: 'Elephant',
    favorite_foods: [ 'grass', 'fruits', 'leaves' ],
    expenses: 150000
  },
  {
    animal_name: 'Giraffe',
    favorite_foods: [ 'leaves', 'twigs', 'fruits' ],
    expenses: 65000
```

mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
    favorite_foods: [ 'meat', 'deer', 'boar' ],
    expenses: 92000
  },
  {
    animal_name: 'Elephant',
    favorite_foods: [ 'grass', 'fruits', 'leaves' ],
    expenses: 150000
  },
  {
    animal_name: 'Giraffe',
    favorite_foods: [ 'leaves', 'twigs', 'fruits' ],
    expenses: 65000
  },
  {
    animal_name: 'Bear',
    favorite_foods: [ 'fish', 'honey', 'berries' ],
    expenses: 78000
  }
]
animal> db.domestic_animals.find(
|   { life_span: { $gt: 5 } },
|   { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
| ).pretty()
[
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'chicken', 'biscuits' ],
    expenses: 25000
  },
  {
    animal_name: 'Cat',
    favorite_foods: [ 'fish', 'milk', 'chicken' ],
    expenses: 18000
  },
  {
    animal_name: 'Cow',
    favorite_foods: [ 'grass', 'hay', 'grains' ],
    expenses: 40000
  },
  {
    animal_name: 'Parrot',
    favorite_foods: [ 'seeds', 'fruits', 'nuts' ],
    expenses: 12000
  },
  {
    animal_name: 'Rabbit',
    favorite_foods: [ 'carrots', 'greens', 'hay' ],
    expenses: 8000
  }
]
animal>
```